

III B.Tech I Semester Regular Examinations, Nov/Dec 2025
Natural Language Processing and Models
(Common to CSE(AIML) & CSE(AI))
(For 2023 Admitted Batch Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) What are the challenges faced in Natural Language Processing?
- 1.b) Define N-gram with an example.
- 1.c) Define Unigram model with an example.
- 1.d) What is the role of Naive Bayes in text classification?
- 1.e) How does LSTM network differ from a GRU network?
- 1.f) What do you understand by Named Entity Recognition (NER)? Give an example.
- 1.g) What are the main components of a transformer architecture?
- 1.h) Expand BERT and state its main goal in language understanding.
- 1.i) Give any two applications of Chatbots.
- 1.j) What is the purpose of Multilingual NLP?

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Explain applications and challenges of Natural Language Processing.
(OR)
b) Explain a complete NLP processing pipeline starting from raw text to feature extraction.
- 3 a) Explain how the Bag of Words and TF-IDF models are used to convert text into numerical features. Use suitable examples to show how the feature vectors are generated.
(OR)
b) Design a text classification system that combines TF-IDF features and Word Embeddings.
- 4 a) Explain how a basic neural network works for NLP tasks. Illustrate the architecture with a neat diagram and suitable example.
(OR)
b) Design a deep learning model for sentiment analysis that uses both CNN and LSTM layers. Explain how combining these two networks helps the model understand and classify text more effectively.
- 5 a) Explain how the attention mechanism works in NLP models with a neat diagram.
(OR)
b) Design a transformer-based model that can handle different NLP tasks like text classification, summarization, and sentiment analysis.
- 6 a) Compare Rule-based and Neural Machine Translation systems.
(OR)
b) Evaluate the real-world impact of Multilingual NLP in improving digital inclusion and accessibility for diverse language communities.

III B.Tech I Semester Regular Examinations, Nov/Dec 2025

Operating Systems & System Programming

(Common to CSE(AIML) & CSE(AI))

(For 2023 Admitted Batch Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) List the basic functions of an operating system
- 1.b) Differentiate between deadlock prevention and avoidance.
- 1.c) Explain fixed partitioning and dynamic partitioning
- 1.d) Explain the main objectives of RAID
- 1.e) What is the role of the I/O subsystem in an operating system?
- 1.f) Explain the difference between security and protection.
- 1.g) What is firmware? How is it different from software?
- 1.h) Differentiate between a linker and a loader
- 1.i) Define the term *syntax* and *semantics*.
- 1.j) What is the purpose of the fork() system call?

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Define a System Call. Explain different types of system calls with suitable examples.
(OR)
 - b) Explain the Dining Philosophers Problem and discuss how it can be solved using Semaphores and Monitors.
- 3 a) What are file access methods? Describe sequential, direct, and indexed access methods in detail.
(OR)
 - b) Describe different RAID levels (RAID 0 to RAID 5) with diagrams and their characteristics.
- 4 a) Describe the layers of I/O software in an operating system. Explain how device drivers communicate between hardware and system software.
(OR)
 - b) What is the difference between discretionary and mandatory access control? Explain with examples.
- 5 a) Discuss different data structures used to implement symbol tables (linear lists, hash tables, trees).
(OR)
 - b) Describe how linking and loading work together to execute a program.
- 6 a) What is the semantic gap? Discuss how compilers help in bridging this gap.
(OR)
 - b) Explain the structure of a file and directory system in UNIX.

III B.Tech I Semester Regular Examinations, Nov/Dec 2025
Computer Vision & Image Processing
(Common to CSE(AIML) & CSE(AI))
(For 2023 Admitted Batch Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) Write the differences between Computer Vision and Image Processing
- 1.b) What is the role of a kernel in spatial filtering?
- 1.c) Write a short note on K-means clustering for image segmentation.
- 1.d) What is region-based image segmentation?
- 1.e) Mention any two depth estimation techniques used in stereo vision.
- 1.f) What are extrinsic parameters in camera calibration?
- 1.g) What is a deformable part model?
- 1.h) State one application of autoencoders in computer vision.
- 1.i) Define dilation and state its effect on binary images.
- 1.j) List any two image compression standards and mention their characteristics.

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Explain the concepts of sampling and quantization in digital image processing with suitable diagrams.
(OR)
b) Explain the concept of Fourier Transform in image processing. How does the Discrete Fourier Transform (DFT) help in frequency analysis of images?
- 3 a) Explain edge detection in image processing. Compare any two popular edge detection techniques.
(OR)
b) What are Gabor filters? Explain their role in transform-based texture analysis. Give examples of their application.
- 4 a) Explain epipolar geometry with a neat diagram. How is it used in stereo vision for correspondence matching?
(OR)
b) Discuss the applications of 3D vision and motion analysis in real-world scenarios such as robotics, autonomous vehicles, and medical imaging.
- 5 a) Explain SIFT and SURF algorithms. How are they used for feature detection and matching in images?
(OR)
b) Compare deep learning architectures (CNN, RNN, Auto-encoders, GANs) used in vision applications.
- 6 a) Discuss the role of various morphological operations with suitable examples and diagrams.
(OR)
b) Explain lossy and lossless image compression techniques. Describe JPEG and PNG standards with their applications.

Review

Code: 23ACD11

R23

III B.Tech I Semester Regular Examinations, Nov/Dec 2025
Data Analysis with Python
(Common to CSE(AIML) & CSE (AI))
(For 2023 Admitted Batch Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) What is the main goal of EDA?
- 1.b) List out the types of dimensions in Numpy
- 1.c) Name two data normalization techniques.
- 1.d) What is scaling in data preprocessing?
- 1.e) What is a uni variate analysis?
- 1.f) Define dispersion and list two measures of dispersion.
- 1.g) Name two Python libraries used for data visualization.
- 1.h) What does an interactive visualization offer over static plots?
- 1.i) Name one real-world dataset suitable for EDA.
- 1.j) Name one method to detect outliers in a dataset.

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Compare the features of Jupyter, Anaconda, and VS Code for EDA.
(OR)
b) Write a summary on the importance of data types and how to check them in Pandas.
- 3 a) Discuss different data scaling and normalization techniques with examples.
(OR)
b) Discuss the significance and implementation of data transformation steps in EDA.
- 4 a) Explain how heat maps are used for correlation analysis.
(OR)
b) Illustrate univariate analysis of a dataset in Python, including all basic plots and summary statistics.
- 5 a) Illustrate advanced visualizations such as violin plots, strip plots, and swarm plots.
(OR)
b) Explain, with code, how to create and interpret composite plots (e.g., grid plots, pair plots) for EDA.
- 6 a) Discuss feature engineering methods in EDA with appropriate examples.
(OR)
b) Illustrate using Python how to identify, visualize, and handle anomalies in real-world datasets.

Code: 23ACS24

R23

III B.Tech I Semester Regular Examinations, Nov/Dec 2025
Introduction to Quantum Technologies and Applications
(Common to All branches)

(For 2023 Admitted Batch Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) State any two postulates of quantum mechanics.
- 1.b) What is the physical meaning of measurement in quantum mechanics?
- 1.c) What is the Hadamard gate used for?
- 1.d) What is Grover's algorithm used for?
- 1.e) What is the purpose of the BB84 protocol?
- 1.f) Define quantum channel.
- 1.g) What is a gravimeter?
- 1.h) Mention one industrial application of quantum sensing.
- 1.i) What are topological insulators?
- 1.j) Mention one quantum initiative in the USA.

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Compare the classical and quantum paradigms with suitable examples.
(OR)
b) Discuss how the principles of quantum mechanics form the foundation of quantum technologies.
- 3 a) Explain the representation of a qubit on the Bloch Sphere.
(OR)
b) Explain how superposition and entanglement enable quantum computation.
- 4 a) Explain the process of quantum teleportation with a neat diagram.
(OR)
b) Briefly explain (a) Quantum channels (b) Quantum internet (c) Quantum security.
- 5 a) Explain the principles of quantum sensing and its advantages.
(OR)
b) Give short notes on (a) Atomic clocks (b) Gravimeters (c) Magnetometers.
- 6 a) Explain the properties and types of quantum materials.
(OR)
b) Give short notes on (a) Superconductors (b) Quantum careers (c) Industry initiatives.

III B.Tech I Semester Regular Examinations, Nov/Dec 2025
Entrepreneurship and New Venture Creation
(Common to All branches)
(For 2023 Admitted Batch Only)

Time: 3 Hours

Max.Marks:70

PART-A

Answer all questions and all questions carry equal marks

10 x 2 = 20M

- 1.a) Mention any two skills essential for Intrapreneurial leadership.
- 1.b) List any two skill sets required for an entrepreneur.
- 1.c) How can socio-economic trends influence the creation of new startups?
- 1.d) What is a customer persona and why is it important for start-ups?
- 1.e) Define prototyping in the context of startups.
- 1.f) How does competition analysis help determine a startup's market position?
- 1.g) What are the 9 blocks of a Lean Canvas model?
- 1.h) What are Debt and Equity in financing?
- 1.i) State any two factors that influence scaling decisions.
- 1.j) Define the term "Venture Pitch."

PART-B

Answer All questions and all questions carry equal marks

5 x 10 = 50M

- 2 a) Explain the concept of entrepreneurship and describe its key features with examples.
(OR)
b) Differentiate between the attributes and mindsets of entrepreneurs and intrapreneurs. How do these differences impact innovation?
- 3 a) Discuss how macro-problem identification helps entrepreneurs understand larger market or societal issues.
(OR)
b) Analyze how demographic, psychographic, and behavioral factors influence customer identification.
- 4 a) Explain how customer "Jobs-to-be-Done" help in identifying real customer needs.
(OR)
b) Analyze the importance of market sizing in determining the overall potential and viability of a business opportunity.
- 5 a) Explain the concept of a business model. Describe different types of business models with suitable examples.
(OR)
b) Why is sales planning important for startups? Explain briefly.
- 6 a) Explain the concept of business scaling and discuss the key factors that influence the scalability of a startup.
(OR)
b) Explain the role of team, traction, and business model in gaining investor confidence during a pitch.